ANALOG NEURAL NETWORK TEST BOARD (70x100mm)

DIY Photolithography & Photoresist PCB Etching Guide - US Letter Format

1. BOARD SPECIFICATIONS & PRINT RULES

* Dimensions: 70.0 mm height x 100.0 mm width (4:3 aspect ratio)
* Substrate: Standard 1.6 mm FR-4 (1 oz / 35 um Copper Clad)
* Page Format: US Letter (8.5" x 11.0" / 215.9 x 279.4 mm) - Landscape
* CRITICAL: Print at 100% SCALE (Actual Size). DO NOT FIT TO PAGE!
* Caliper Check: Verify 50.0 mm scale bar with digital calipers.

2. PHOTORESIST PAGE DIRECTORY

Page 2: Front Copper (F.Cu) - POSITIVE RESIST (Normal View)
Page 3: Front Copper (F.Cu) - POSITIVE RESIST (Mirrored View - Emulsion Down)
Page 4: Back Copper (B.Cu) - POSITIVE RESIST (Normal View)
Page 5: Back Copper (B.Cu) - POSITIVE RESIST (Mirrored View)
Page 6: Front Copper (F.Cu) - NEGATIVE DRY FILM (Normal View)
Page 7: Front Copper (F.Cu) - NEGATIVE DRY FILM (Mirrored View - Emulsion Down)
Page 8: Back Copper (B.Cu) - NEGATIVE DRY FILM (Normal View)
Page 9: Back Copper (B.Cu) - NEGATIVE DRY FILM (Mirrored View)
Page 10: Component Silkscreen, Drilling Map & Reference Legend

3. CHEMICAL PROCESSES & RECIPES

A. Positive Pre-Sensitized Boards (Bungard / Positiv 20):

* Artwork: Solid black traces block UV (Pages 2 - 5).
* Developer: 10 g / L Sodium Hydroxide (NaOH) in water at 20 C.
* Dev Time: 60 - 90 seconds until unexposed resist clears.

B. Negative Dry Film Photoresist (Riston Blue Film):

* Artwork: White traces transmit UV to cure (Pages 6 - 9).
* Developer: 10 g / L Sodium Carbonate (Na2CO3 / Washing Soda).
* Dev Time: 90 - 120 seconds with gentle agitation.

C. Copper Etching Baths:

* Option 1: Ferric Chloride (FeCl3) at 45 - 50 C with aeration.
* Option 2: Cupric Chloride (CuCl2 + HCl) - regenerative.
* Option 3: 2 parts Hydrogen Peroxide (3%) + 1 part Muriatic Acid.

4. DRILLING & VIAS GUIDE

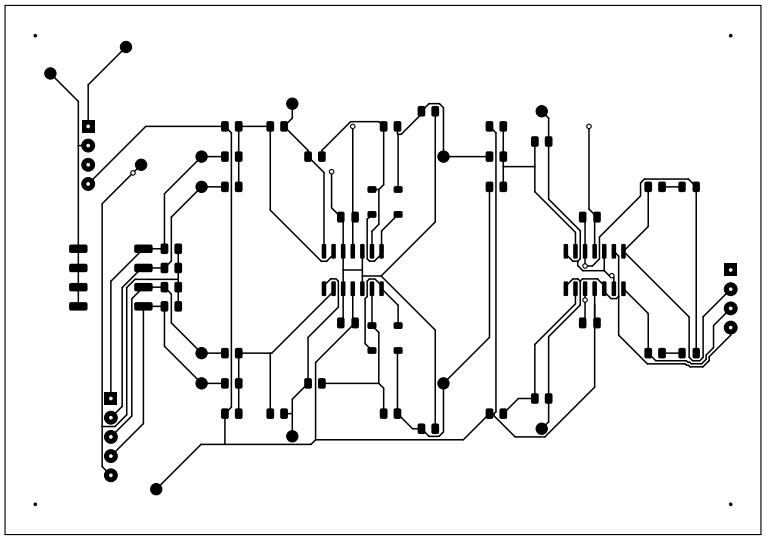
* Pad center marks indicate optical center for drill bits.
* Vias: 0.80 mm carbide bit (solder with thin copper wire through-hole).
* Header Pins (J_IN, J_PWR, J_OUT): 1.00 mm bit.
* Corner Mounting Holes (H1..H4): 3.20 mm bit (M3 hardware).

PRO-TIP: OPTICAL CONTACT & EMULSION-SIDE ORIENTATION

To eliminate light undercut (diffraction around traces), the printed toner/ink emulsion side MUST touch the copper clad directly. Use MIRRORED masks when placing toner face-down on the board. The text annotations will read right-side forward through the transparent film, visually confirming 100% correct optical contact before turning on your UV exposure unit.

ANALOG NEURAL NETWORK TEST BOARD (70x100mm) — PHOTORESIST MASK

FORMAT: US LETTER (8.5x11 in) | SCALE: 1:1 TRUE SIZE | ALL-IN-ONE 4-IN / 2-HIDDEN / 2-OUT WTA



50.0 mm [1:1 SCALE CALIBRATION BAR — MEASURE WITH CALIPERS]

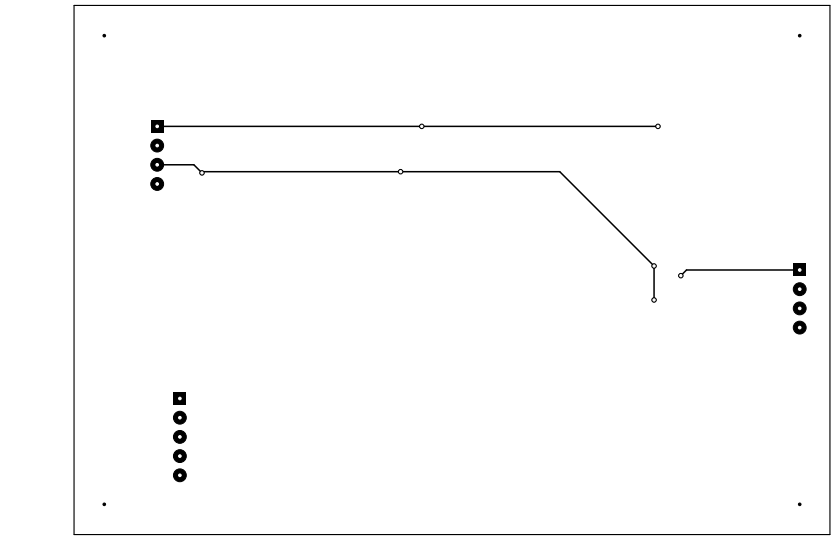
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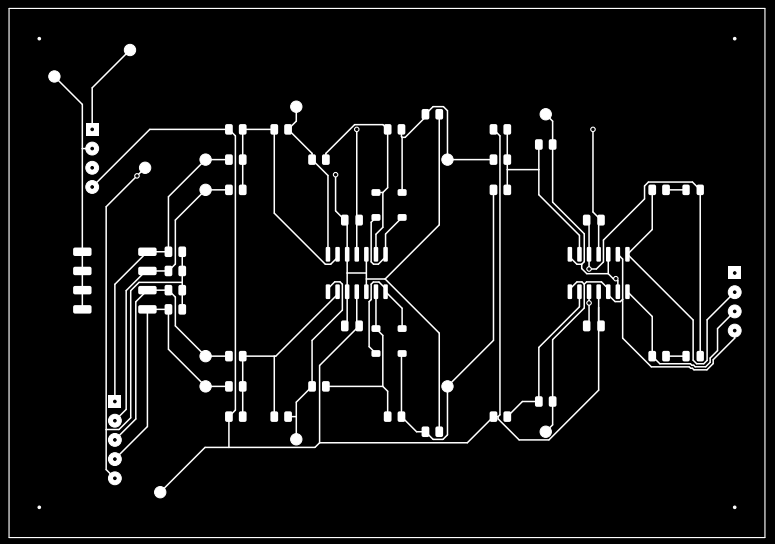
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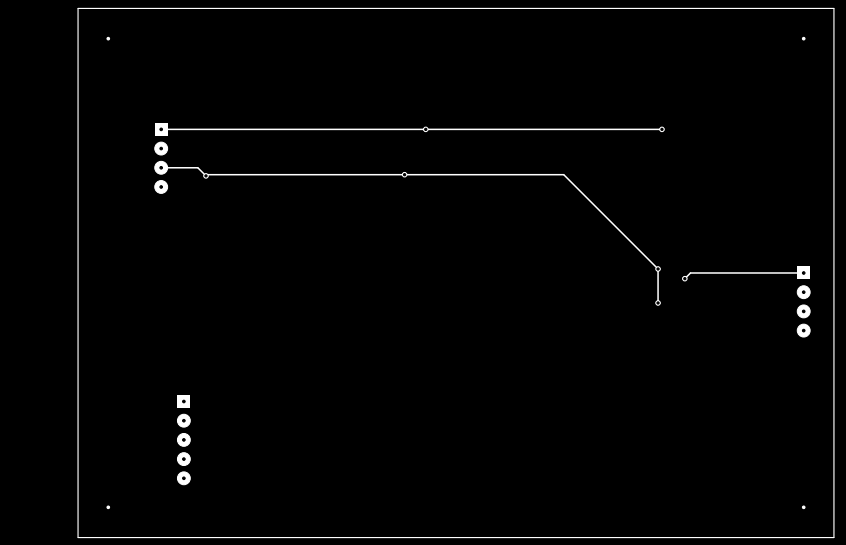
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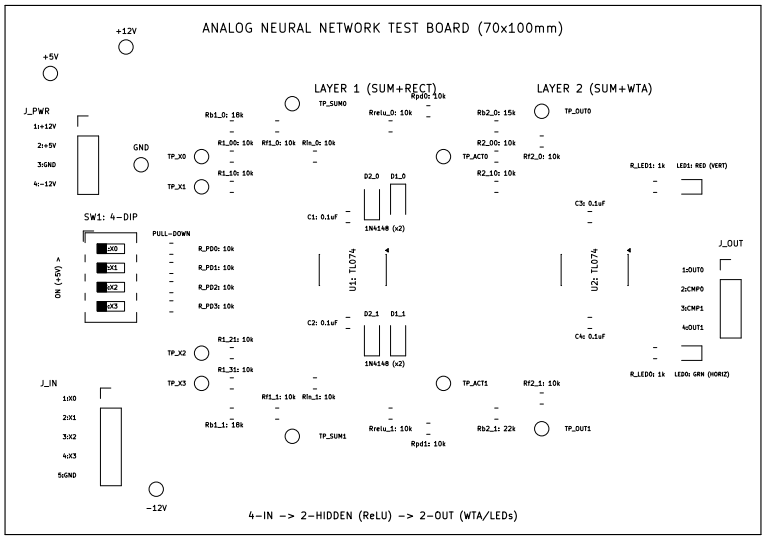


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ANALOG NEURAL NETWORK TEST BOARD (70x100mm)
+12V
+5V
J_PWR
1:+12V
2:+5V
3:GND
4:-12V
GND
SW1: 4-DIP
ON (+5V)
PULL-DOWN
X0
X1
X2
X3
R_PD0: 10k
R_PD1: 10k
R_PD2: 10k
R_PD3: 10k
J_IN
1:X0
2:X1
3:X2
4:X3
5:GND
-12V
LAYER 1 (SUM+RECT)
LAYER 2 (SUM+WTA)
TP_X0
TP_X1
TP_X2
TP_X3
Rb1_0: 18k
R1_00: 10k
R1_10: 10k
Rb1_1: 18k
R1_21: 10k
R1_31: 10k
Rf1_0: 10k
Rf1_1: 10k
Rin_0: 10k
Rin_1: 10k
TP_SUM0
TP_SUM1
Rrelu_0: 10k
Rrelu_1: 10k
Rpd0: 10k
Rpd1: 10k
D2_0
D1_0
D2_1
D1_1
1N4148 (x2)
1N4148 (x2)
C1: 0.1uF
C2: 0.1uF
U1: TL074
Rb2_0: 15k
R2_00: 10k
R2_10: 10k
Rb2_1: 22k
Rf2_0: 10k
Rf2_1: 10k
TP_ACT0
TP_ACT1
TP_OUT0
TP_OUT1
C3: 0.1uF
C4: 0.1uF
U2: TL074
R_LED1: 1k
R_LED0: 1k
LED1: RED (VERT)
LED0: GRN (HORIZ)
J_OUT
1:OUT0
2:CMP0
3:CMP1
4:OUT1
4-IN -> 2-HIDDEN (ReLU) -> 2-OUT (WTA/LEDs)